

MAT 344 HW02

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Problem 1. Show for a complex valued function f that

$$df = \frac{\partial f}{\partial z} dz + \frac{\partial f}{\partial \bar{z}} d\bar{z}.$$

Solution. Recalling that

$$\frac{\partial f}{\partial z} = \frac{1}{2} \left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right), \quad \frac{\partial f}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right),$$

we directly compute

$$\begin{aligned} \frac{\partial f}{\partial z} dz + \frac{\partial f}{\partial \bar{z}} d\bar{z} &= \frac{1}{2} \left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right) \cdot (dx + i dy) + \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) \cdot (dx - i dy) \\ &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = df. \end{aligned} \quad \square$$

Problem 2. For which values of z is

$$\sum_{n=0}^{\infty} \left(\frac{z}{1+z} \right)^n$$

convergent?

Solution. A geometric series of complex numbers converges if and only if the common ratio of the sequence being summed has magnitude smaller than 1, so we want to find the subset of $\mathbb{C}$ containing all z such that $|\frac{z}{1+z}| < 1$. Letting $z = a + bi$ for real a, b , this is equivalent to

$$\sqrt{a^2 + b^2} < \sqrt{(1+a)^2 + b^2} \iff a^2 < (1+a)^2 \iff -\frac{1}{2} < a,$$

i.e. the series converges for $\boxed{\text{all } z \text{ such that } \operatorname{Re}(z) > -\frac{1}{2}}$. $\square$

Problem 3. Using the skew-symmetric rules for wedge multiplication of real 1-forms, show that

$$dz \wedge dz = d\bar{z} \wedge d\bar{z} = 0 \quad \text{and} \quad dz \wedge d\bar{z} = -d\bar{z} \wedge dz = -2i dx \wedge dy.$$

Use this to express the formula in Green's theorem for $\omega = f(z)dz$ (where $f(x, y) = u(x, y) + iv(x, y)$) in terms of $\frac{\partial f}{\partial \bar{z}} dz \wedge d\bar{z}$.

Solution. Wedging a 1-form with itself is always zero because swapping the arguments yields $\omega \wedge \omega = -\omega \wedge \omega$ by anticommutativity. This implies that $2\omega \wedge \omega = \omega \wedge \omega = 0$, so in particular $dz \wedge dz = d\bar{z} \wedge d\bar{z} = 0$. The other two equations follow from a direct calculation:

$$dz \wedge d\bar{z} = (dx + i dy)(dx - i dy) = dx \wedge dx + dy \wedge dy + i(dy \wedge dx - dx \wedge dy) = -2i dx \wedge dy,$$

and this is equal to $-d\bar{z} \wedge dz$ by anticommutativity. Green's theorem says that, for a good domain U ,

$$\int_{\partial U} \omega = \int_U d\omega = \int_U df \wedge dz = \int_U \left(\frac{\partial f}{\partial z} dz + \frac{\partial f}{\partial \bar{z}} d\bar{z} \right) \wedge dz = \int_U \frac{\partial f}{\partial \bar{z}} dz \wedge d\bar{z},$$

where the other summand in the integral cancels out because $dz \wedge dz = 0$. $\square$

Problem 4. Let $\Delta = \{z \in \mathbb{C} : |z| \leq 1\}$ and choose $a \in \text{Int } \Delta$. Let $\varepsilon > 0$ be small enough that $\Delta(\varepsilon, a) \equiv \{z \in \mathbb{C} : |z - a| < \varepsilon\} \subseteq \text{Int } \Delta$. Let f be a holomorphic function in a neighborhood of Δ . Write out Green's formula for the holomorphic function

$$F(z) = \frac{f(z)}{z - a}$$

on $\Omega \equiv \Delta - \text{Int } \Delta(\varepsilon, a)$. What happens as $\varepsilon \rightarrow 0$?

Solution. Let $g(z) = \frac{f(z)}{z - a}$. Since $a \notin \Omega$, $\frac{1}{z - a}$ is holomorphic on Ω , hence so is g , being the product of two functions that are holomorphic on Ω . In particular, $\frac{\partial g}{\partial \bar{z}} = 0$ on Ω , so

$$\int_{\partial\Omega} g(z) dz = \int_{\Omega} \frac{\partial g}{\partial \bar{z}} dz \wedge d\bar{z} = 0$$

by the result of Problem 3. If we orient the boundaries of Δ and $\Delta(\varepsilon, a)$ counterclockwise, the left-hand side equalling zero implies

$$\int_{\partial\Delta} g(z) dz = \int_{\partial\Delta(\varepsilon, a)} g(z) dz. \quad (*)$$

Let $\gamma_r = \partial\Delta(r, a)$, which we can parametrize as the curve $\gamma_r : [0, 2\pi] \rightarrow \mathbb{C}$ given by $\gamma_r(\theta) = a + re^{i\theta}$. Then,

$$\int_{\gamma_r} \frac{f(z)}{z - a} dz = \int_0^{2\pi} \frac{f(a + re^{i\theta})}{re^{i\theta}} \cdot rie^{i\theta} d\theta = \int_0^{2\pi} if(a + re^{i\theta}) d\theta.$$

Since f is holomorphic at a , there exists a constant $C > 0$ such that, for each $r > 0$, there is a $\delta > 0$ satisfying $|f(a + re^{i\theta}) - f(a)| < C|re^{i\theta}| = Cr$ (in particular, any $C > |f'(a)|$ has this property). This implies that

$$\left| \int_0^{2\pi} if(a + re^{i\theta}) d\theta - 2\pi if(a) \right| = \left| \int_0^{2\pi} i(f(a + re^{i\theta}) - f(a)) d\theta \right| < \int_0^{2\pi} Cr d\theta = 2\pi Cr,$$

so, in the limit,

$$\lim_{r \rightarrow 0} \int_{\gamma_r} \frac{f(z)}{z - a} dz = f(a).$$

By (*), the argument of the limit is constant as r varies, so in fact this implies

$$\int_{\partial\Delta} \frac{f(z)}{z - a} dz = \int_{\gamma_r} \frac{f(z)}{z - a} dz = f(a). \quad \square$$